

Total marks - 120

Attempt Questions 1-8

All questions are of equal value

Answer each question on a NEW PAGE.

Marks

Question 1 (15 marks) Start a NEW page.

(a) Find $\int \frac{\cos^3 x}{\sin^2 x} dx$. **3**

(b) Find $\int \operatorname{cosec} x \, dx$. **3**

(c) Find $\int \frac{2x-1}{x^2-6x+10} dx$. **4**

(d) Let $I_n = \int_1^e x(\ln x)^n dx$ for $n = 0, 1, 2, 3, \dots$ **5**

(i) Show that $I_n = \frac{e^2}{2} - \frac{n}{2} I_{n-1}$.

(ii) Hence find $\int_1^e x(\ln x)^2 dx$.

Question 2 (15 marks) Start a NEW page.

- (a) If $z = -1 + i\sqrt{3}$, express each of the following in the form $a+ib$ where a and b are real 5
- (i) $\bar{z}$.
- (ii) z^2 .
- (iii) $\frac{1}{z}$.
- (iv) z^6 .
- (b) Given $z_1 = -1 - i$ and $z_2 = 3 + i$, draw neat labeled sketches to show the locus of z where: 5
- (i) $|z - z_1| \leq |z - z_2|$.
- (ii) $0 \leq \arg(z - z_1) \leq \frac{\pi}{4}$.
- (iii) $\arg(z - z_1) = \arg(z - z_2)$.
- (c) The complex number z satisfies the equation $z\bar{z} + 2iz = 12 + 6i$. Find all possible values of z . 3
- (d) The quadratic equation $z^2 - (1 + i)z + 2i = 0$ has roots α, β . Find, in simplest form, the value of $\alpha^{-2} + \beta^{-2}$. 2

Question 3 (15 marks) Start a NEW page.

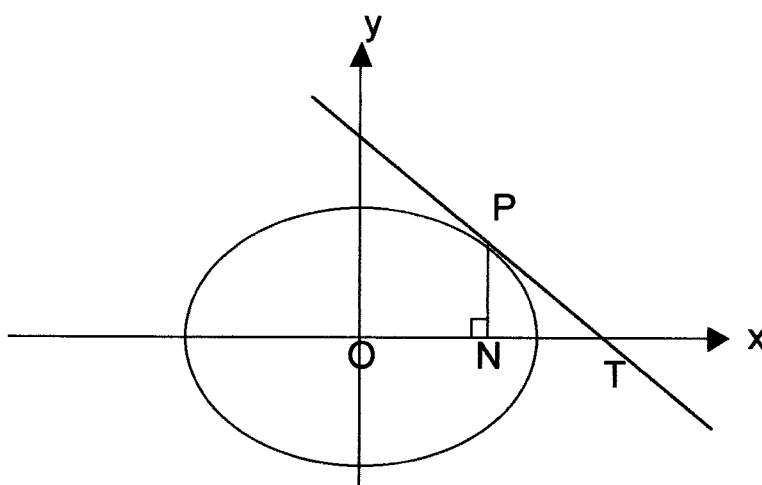
- (a) For the conic $9x^2 - 16y^2 = 144$, sketch the curve, showing foci, directrices and asymptotes.

5

- (b) The tangent at $P(a \cos \theta, b \sin \theta)$ on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ cuts the x axis at T . The perpendicular PN is drawn to the x axis.

Prove that $ON \cdot OT = a^2$.

5



Question 3 continues on page 5

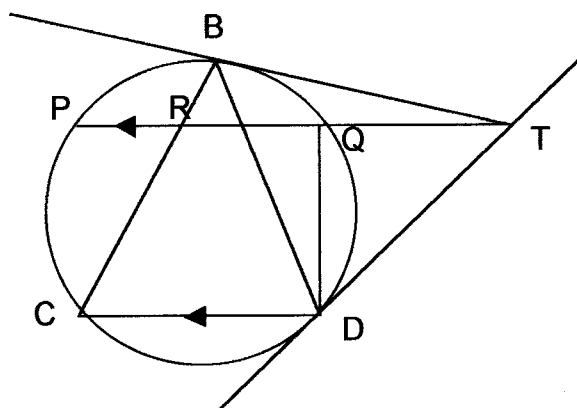
Question 3 continued

- (c) PQ and CD are parallel chords of a circle.
 The tangent at D cuts PQ extended at T.
 B is the point of contact of the other tangent from T to the circle.
 BC meets PQ at R.

5

Copy the diagram onto your answer sheet.

- (i) Prove that $\angle BDT = \angle BRT$.
 (ii) Prove that B, T, D, R are concyclic.
 (iii) Prove that $\angle BRT = \angle DRT$.



Question 4 (15 marks) Start a NEW page.

(a) The cubic equation $x^3 - 2x^2 - 3x - 4 = 0$ has roots α, β, γ **5**

(i) Find the value of $\frac{\alpha}{\beta\gamma} + \frac{\beta}{\alpha\gamma} + \frac{\gamma}{\alpha\beta}$

(ii) Form the equation with integer coefficients whose roots are $\alpha^{-1}, \beta^{-1}, \gamma^{-1}$

(b) The polynomial $P(z) = z^4 - 2z^3 - 7z^2 + 26z - 20$ has a zero at $z = 2 + i$. Find all of the zeros of $P(z)$ **4**

(c) $P\left(cp, \frac{c}{p}\right)$, $p > 0$, and $Q\left(cq, \frac{c}{q}\right)$, $q > 0$, are two points on the rectangular hyperbola $xy = c^2$. The tangents at P and Q intersect at R. Given that the equation of chord PQ is $x + pqy = c(p + q)$. **6**

(i) Find the equation of the tangent at P.

(ii) Find the coordinates of R.

(iii) If the secant PQ passes through $(3c, 0)$, find the locus of R and state any restrictions on the locus.

Question 5 (15 marks) Start a NEW page.

- (a) The region bounded by the curve $y = 2x - x^2$ is rotated about the line $x = 2$ to form a solid of revolution. By taking slices perpendicular to the line $x = 2$, find the volume of the solid. 5
- (b) The Great Pyramid of Cheops is approximately 150 metres high and its base is a square of approximate area 5 hectares. 4
- (i) Show that the area of the cross-section of a square pyramid at height y metres above the base is given by $A(y) = \left(\frac{h-y}{h}\right)^2 \times A$, where A is the area of the base, and h is the height of the pyramid.
- (ii) Use the slice technique to find the volume of the Great Pyramid of Cheops
- (c) On a certain day, the depth of water in a harbour at high tide at 5 am is 9 metres. At the following low tide at 11.20 am the depth is 3 metres. Find the latest time before noon that a ship can enter the harbour if a minimum depth of 7.5 metres of water is required. Assume that the tides are undergoing simple harmonic motion. 6

Question 6 (15 marks) Start a NEW page.

- (a) (i) Let OABC be a square on the Argand diagram where O is the origin and neither A nor C is on the axes. The points A and C represent the complex numbers z and iz respectively.

Show that B represents the complex number $z(1+i)$

4

- (ii) The square is now rotated about O through 45° in an anticlockwise direction to OA'B'C'.

Find the complex number represented by the point B'.

- (b) (i) Show that $\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \cos x + \sin x} = \ln 2$

5

- (ii) Hence evaluate $\int_0^{\frac{\pi}{2}} \frac{x}{1 + \cos x + \sin x} dx$, using the substitution $u = \frac{\pi}{2} - x$

- (c) If $f(x) = (x-1)(x-3)$ sketch the following curves, showing all intercepts, asymptotes and turning points. Draw a separate graph for each.

6

(i) $y = \frac{1}{f(x)}$

(ii) $y = [f(x)]^2$

(iii) $y^2 = f(x)$

(iv) $|y| = f(x)$

Question 7 (15 marks) Start a NEW page.

- (a) (i) Sketch $y = \frac{1}{x^2 + 1}$ and $y = \frac{x^2}{x^2 + 1}$ showing the coordinates of their points of intersection.

9

- (ii) The region bounded by these curves is rotated about the y axis to form a solid of revolution. By considering the solid as the sum of cylindrical shells, find the volume.

- (b) The Fibonacci Sequence, F_n , is defined by :

$$F_1 = 1 \quad F_2 = 1 \quad F_{n+2} = F_{n+1} + F_n, \text{ for all } n \geq 1$$

6

- (i) Prove that $F_8 = 3 \times F_5 + 2 \times F_4$
- (ii) Prove, by mathematical induction, that F_{4n} is divisible by 3, for all positive integers n .

Question 8 (15 marks) Start a NEW page.

- (a) (i) Use DeMoivre's theorem to show that $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$
- (ii) Use this result to solve the equation $8x^3 - 6x + 1 = 0$ 5
- (iii) Deduce that $\sec \frac{\pi}{9} \sec \frac{2\pi}{9} \sec \frac{4\pi}{9} = 8$
- b) A curve has parametric equations $x = \theta - \sin \theta$ and $y = 1 - \cos \theta$ 10
- (i) Show that $\frac{dy}{dx} = \cot \frac{\theta}{2}$
- (ii) Hence show that $\frac{d^2y}{dx^2} = -\frac{1}{y^2}, y \neq 0$
- (iii) Show that the curve has stationary points at $(n\pi, 2)$, for n odd.
- (iv) Determine the nature of these stationary points.
- (v) Sketch the curve, showing stationary points and intercepts on the axes.
- (vi) Discuss the nature of the points at which the curve intersects the x axis.

End of the paper

Question 1

$$a \int \frac{\cos^3 x}{\sin^2 x} dx = \int \frac{(1-\sin^2 x) \cos x}{\sin^2 x} dx$$

1 mark

$$\text{let } u = \sin x \\ du = \cos x dx$$

$$= \int (u^{-2} - 1) du$$

1 mark

$$= -\frac{1}{u} - u$$

$$= -\operatorname{cosec} x - \sin x + c$$

1 mark

$$b \int \operatorname{cosec} x dx = \int \frac{dx}{\sin x}$$

$$t = \tan \frac{x}{2}$$

$$dx = \frac{2dt}{1+t^2}$$

$$\sin x = \frac{2t}{1+t^2}$$

$$= \int \frac{2dt}{\frac{1+t^2}{\frac{2t}{1+t^2}}}$$

1 mark

$$= \int \frac{dt}{t}$$

1 mark

$$= \ln |t|$$

$$= \ln \left| \tan \frac{x}{2} \right| + c$$

1 mark.

$$\underline{\text{OR}} \quad -\ln |\operatorname{cosec} x + \cot x| + c$$

$$\underline{\text{OR}} \quad \ln |\operatorname{cosec} x - \cot x| + c$$

$$c \int \frac{2x+1}{x^2-6x+10} dx = \int \frac{2x-6}{x^2-6x+10} dx + \int \frac{7}{x^2-6x+10} dx$$

1 mark

$$= \ln |x^2-6x+10| + 7 \int \frac{dx}{(x-3)^2+1}$$

2 marks

$$= \ln |x^2-6x+10| + 7 \tan^{-1}(x-3) + c$$

1 mark.

$$d \text{ (i) } I_n = \int_1^e x (\ln x)^n dx$$

$$u = (\ln x)^n \quad v = \frac{x^2}{2}$$

$$u' = \frac{n}{x} (\ln x)^{n-1} \quad v' = x$$

1 mark $\begin{matrix} u & v \\ u' & v' \end{matrix}$

$$\therefore I_n = \left[\frac{x^2}{2} (\ln x)^n \right]_1^e - \frac{n}{2} \int_1^e x (\ln x)^{n-1} dx$$

1 mark

$$= \frac{e^2}{2} - \frac{n}{2} I_{n-1}$$

1 mark.

$$\text{(ii) } \int_1^e x (\ln x)^2 dx = I_2 = \frac{e^2}{2} - I_1 \\ = \frac{e^2}{2} - \left(\frac{e^2}{2} - \frac{1}{2} I_0 \right)$$

1 mark.

$$= \frac{1}{2} \int_1^e x dx$$

$$= \frac{1}{2} \left[\frac{x^2}{2} \right]_1^e$$

$$= \frac{1}{4} (e^2 - 1)$$

1 mark

Question 2

a (i) $\bar{z} = -1 - i\sqrt{3}$ 1 mark

(ii) $z^2 = -2 - 2\sqrt{3}i$ 1 mark

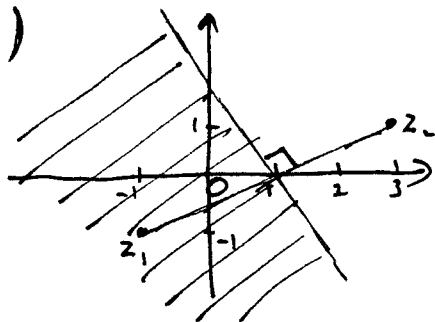
(iii) $\frac{1}{z} = -\frac{1}{4} - \frac{\sqrt{3}}{4}i$ 1 mark

(iv) $z^6 = (2 \operatorname{cis} \frac{2\pi}{3})^6$ 1 mark

$= 64 \operatorname{cis} 4\pi$
 $= 64$

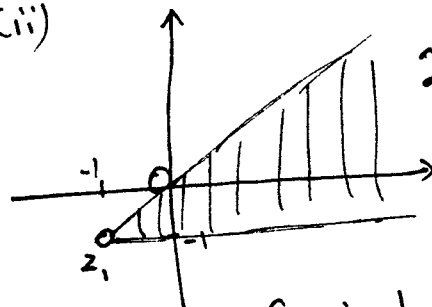
1 mark.

b (i)



1 mark

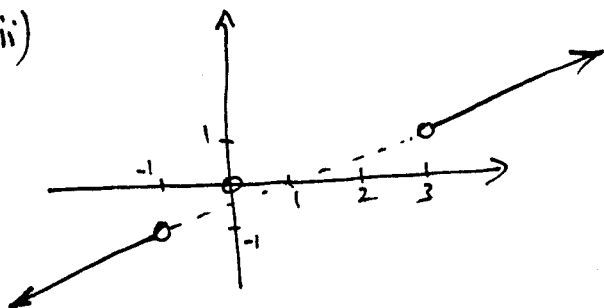
(ii)



2 marks

(lose 1 mark if z_1 included)

(iii)



2 marks

(given 1 mark if interval z_1, z_2 shown)
 (lose 1 mark if z_1, z_2 included).

c Let $z = x + iy$

$\therefore x^2 + y^2 + 2ix - 2y = 12 + 6i$ 1 mark

Equating real & imaginary parts $\Rightarrow x = 3$

$\therefore 9 + y^2 - 2y = 12$

$y = 3, -1$

$\therefore z = 3 + 3i$ or $z = 3 - i$

1 mark for equating real, imaginary parts

1 mark.

d $z^2 - (1+i)z + 2i = 0$

$\therefore \alpha + \beta = 1+i$

$\alpha\beta = 2i$

Now $\alpha^{-2} + \beta^{-2} = \frac{1}{\alpha^2} + \frac{1}{\beta^2}$

$= \frac{\alpha^2 + \beta^2}{(\alpha\beta)^2}$

$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{(\alpha\beta)^2}$

$= \frac{(1+i)^2 - 4i}{-4}$

$= \frac{i}{2}$

1 mark.

Question 3

a $9x^2 - 16y^2 = 144$

$$\frac{x^2}{16} - \frac{y^2}{9} = 1$$

$a=4, b=3$ $q=16(e^2-1)$
 $e = \frac{5}{4}$

1 mark.

Foci $(\pm 5, 0)$

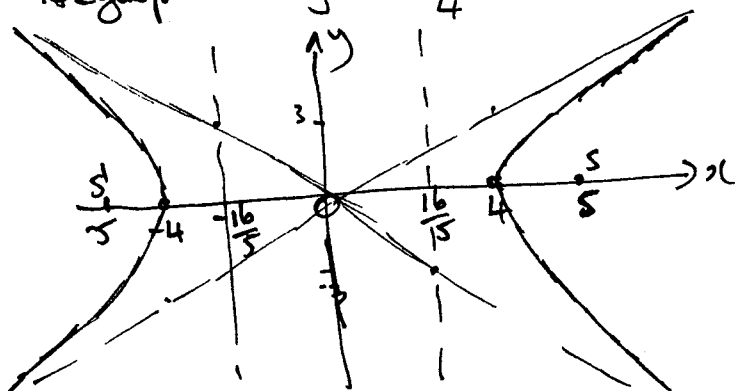
1 mark

Directrices $x = \pm \frac{16}{5}$

1 mark

Asymptotes $y = \pm \frac{3x}{4}$

1 mark.



1 mark.

b $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\frac{2x}{a^2} + \frac{2y}{b^2} y' = 0$$

$$y' = -\frac{b^2 x}{a^2 y}$$

1 mark

At P, grad of tangent = $-\frac{b^2 a \cos \theta}{a^2 b \sin \theta}$
 $= -\frac{b \cos \theta}{a \sin \theta}$

Eqn of tangent at P is $y - b \sin \theta = \frac{-b \cos \theta}{a \sin \theta} (x - a \cos \theta)$ 1 mark.

At T, $y=0 \Rightarrow -a b \sin^2 \theta = -b \cos \theta (x - a \cos \theta)$
 $x - a \cos \theta = \frac{a \sin^2 \theta}{\cos \theta}$
 $x = a \cos \theta + \frac{a \sin^2 \theta}{\cos \theta}$

$$= \frac{a}{\cos \theta} (\cos^2 \theta + \sin^2 \theta)$$

$$= \frac{a}{\cos \theta}$$

$\therefore T$ is $(\frac{a}{\cos \theta}, 0)$

1 mark.

N is $(a \cos \theta, 0)$

1 mark.

1 mark.

Thus $ON \cdot OT = a^2$

$\therefore$ (i) $\angle BDT = \angle BCD$ (angle in alternate segment) 1 mark
 $\angle BCD = \angle BRT$ (corresponding angles in // lines) 1 mark

$\therefore \angle BDT = \angle BRT$

(ii) B, T, D, R are concyclic (BT subtends equal angles at R, D on same side of it) 1 mark.

(iii) $TB = TD$ (tangents from external point equal) 1 mark

$\therefore \angle BRT = \angle DRT$ (equal chords subtend equal angles at circumference) 1 mark.

Question 4

a. $x^3 - 2x^2 - 3x - 4 = 0$

(i) $\alpha + \beta + \gamma = 2$

$\alpha\beta + \alpha\gamma + \beta\gamma = -3$

$\alpha\beta\gamma = 4$

{ 1 mark

Now $\frac{\alpha}{\beta\gamma} + \frac{\beta}{\alpha\gamma} + \frac{\gamma}{\alpha\beta} = \frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha\beta\gamma}$ 1 mark

$= \frac{(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)}{\alpha\beta\gamma}$ 1 mark

$= \frac{4 + 6}{4}$

$= \frac{5}{2}$ 1 mark.

(ii) Required eqn is $(\frac{1}{x})^3 - 2(\frac{1}{x})^2 - 3(\frac{1}{x}) - 4 = 0$

$1 - 2x - 3x^2 - 4x^3 = 0$

$4x^3 + 3x^2 + 2x - 1 = 0$ 1 mark.

b. $2+i$ is a root, real coefficients. $\therefore 2-i$ is a root

Together they form the quadratic factor $z^2 - 4z + 5$

$\begin{array}{r} z^2 - 4z + 5 \\ \underline{z^2 + 2z - 4} \\ 2^4 - 2z^3 - 7z^2 + 26z - 20 \end{array}$

Now $z^2 + 2z - 4 = 0$ has roots $z = -1 \pm \sqrt{5}$

$\therefore$ the 4 zeros are $2 \pm i, -1 \pm \sqrt{5}$

c. (i) Chord PQ given $x + pyq = c(p+q)$

As $q \rightarrow p$, eqn tangent at P is $x + p^2y = 2cp$ 1 mark.

(ii) $x + p^2y = 2cp \dots \textcircled{1}$

$x + q^2y = 2cq \dots \textcircled{2}$

$\textcircled{1} - \textcircled{2} \quad (p^2 - q^2)y = 2c(p - q)$

$y = \frac{2c}{p+q}$ 1 mark

Sub. into $\textcircled{1} \quad x = \frac{2cpq}{p+q}$ 1 mark.

$\therefore R$ is $(\frac{2cpq}{p+q}, \frac{2c}{p+q})$

(iii) PQ through $(3c, 0)$ $\therefore 3c = c(p+q)$

$p+q = 3$

Hence, at R, since $y = \frac{2c}{p+q}$, then $y = \frac{2c}{3}$ is the locus of R 1 mark.

with the restriction that R is in the first quadrant (as $p, q > 0$) and that R is between the hyperbola and the coordinate axes.

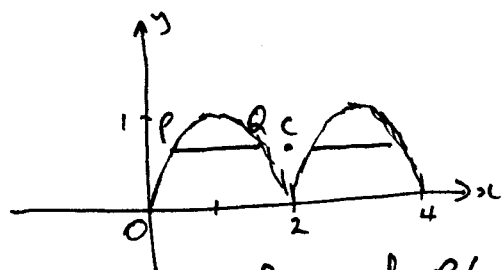
Now $y = \frac{2c}{3}$ cuts $xy = c^2$ when $x = \frac{3c}{2}$

$\therefore$ restriction is $0 < x < \frac{3c}{2}$

{ 1 mark for $x > 0$
1 mark for $x < \frac{3c}{2}$

Question 5

a



$$y = 2x - x^2$$

$$x^2 - 2x + y = 0$$

$$x = \frac{2 \pm \sqrt{4 - 4y}}{2}$$

$$x = 1 \pm \sqrt{1 - y}$$

For P as shown $x = 1 - \sqrt{1 - y}$

Take slice through P(x, y) on $y = 2x - x^2$, thickness δy



$$CP = 2 - x$$

$$CQ = x$$

Area of cross-section is $A = \pi [(2-x)^2 - x^2]$

$$= \pi (4 - 4x)$$

Vol. of slice $\delta V = \pi (4 - 4x) \delta y$

Volume = $\lim_{\delta y \rightarrow 0} \sum_{y=0}^1 \pi (4 - 4x) \delta y$

$$= 4\pi \int_0^1 (1-x) dy$$

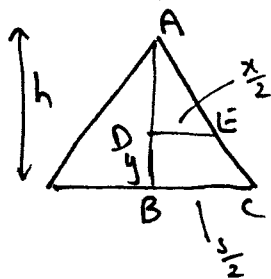
$$= 4\pi \int_0^1 (1-y)^{\frac{1}{2}} dy$$

$$= 4\pi \left[-\frac{2}{3} (1-y)^{\frac{3}{2}} \right]_0^1$$

$$= \frac{8\pi}{3} \text{ cu. units}$$

Marking Correct radii of cross-section 1 mark
 Correct area of cross-section 1 mark.
 $V = 4\pi \int_0^1 (1-x) dy$ 1 mark.
 Obtaining x as function of y 1 mark.
 Correct answer 1 mark.

b (i) Let base of square pyramid be S units square
 & let x be side length of square cross-section at height y



$$\triangle ADE \parallel \triangle ABC$$

$$\therefore \frac{\frac{x}{2}}{\frac{s}{2}} = \frac{h-y}{h}$$

$$x = \frac{(h-y)}{h} \cdot s$$

$\therefore$ Area of square at height y is $\left(\frac{h-y}{h} \right)^2 \cdot s^2$

$$= \left(\frac{h-y}{h} \right)^2 \cdot A$$

1 mark

(ii) Vol. slice $\delta V = \left(\frac{h-y}{h} \right)^2 \cdot A \cdot \delta y$

Total volume $V = \int_0^h \left(\frac{h-y}{h} \right)^2 A dy$

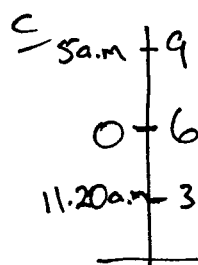
$$= \frac{50000}{150^2} \int_0^{150} (150-y)^2 dy$$

$$= 2500000 \text{ m}^3$$

1 mark

1 mark

1 mark.



Take 0 at 6m level as centre of motion

$$\therefore a = 3$$

1 mark

Take $t=0$ at 5 a.m.

$$\text{Period } T = 12\text{hr } 40\text{min} \\ = \frac{38}{3} \text{ hrs.}$$

$$\therefore n = \frac{3\pi}{19}$$

1 mark

$$\text{Now } x = a \cos(nt + \alpha)$$

$$x = 3 \cos\left(\frac{3\pi t}{19} + \alpha\right)$$

$$\text{But } x=3 \text{ when } t=0 \quad \therefore 3 = 3 \cos \alpha \\ \alpha = 0$$

1 mark for
proving $\alpha=0$

$$\text{Using } x = 3 \cos\left(\frac{3\pi t}{19}\right)$$

require t when $x=1.5$

$$\cos \frac{3\pi t}{19} = \frac{1}{2}$$

1 mark.

$$\frac{3\pi t}{19} = \frac{\pi}{3}, \frac{5\pi}{3}, \dots$$

$$t = \frac{19}{9}, \frac{95}{9}, \dots$$

$$= 2\frac{1}{9} \text{ hrs}, 10\frac{5}{9} \text{ hrs}, \dots$$

1 mark

Thus depth is 7.5m or more between 5 a.m. and 7.07 a.m.

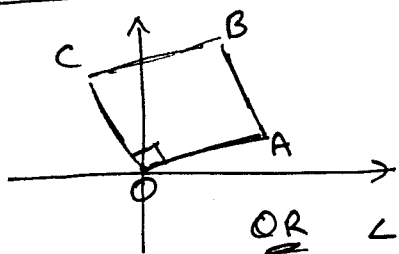
and then from 3.33 p.m. until . . .

$\therefore$ Latest time before noon is 7.07 a.m.

1 mark

Question 6

a (i)



$$\begin{aligned}\vec{OB} &= \vec{OA} + \vec{AB} \\ &= \vec{OA} + \vec{OC}\end{aligned}$$

1 mark

$$\therefore B \text{ represents } 2 + i2 = 2(1+i)$$

1 mark

OR $\angle BOA = 45^\circ$
 $BO = \sqrt{2} \times AO$

} 1 mark

$$\begin{aligned}\therefore B \text{ represents } & 2 \times \sqrt{2} \operatorname{cis} \frac{\pi}{4} \\ &= 2 \times \sqrt{2} \times \frac{1}{\sqrt{2}} (1+i) \\ &= 2(1+i)\end{aligned}$$

1 mark.

$$\begin{aligned}\text{(ii) } B' \text{ represents } & 2(1+i) \operatorname{cis} \frac{\pi}{4} = 2(1+i) \frac{1}{\sqrt{2}} (1+i) \\ &= \frac{2}{\sqrt{2}} (1+i)^2 \\ &= \sqrt{2} 2i\end{aligned}$$

1 mark

1 mark

b (i) $\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \cos x + \sin x}$

Let $t = \tan \frac{x}{2}$

$$= \int_0^1 \frac{\frac{2dt}{1+t^2}}{1 + \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}}$$

1 mark

$$= \int_0^1 \frac{2dt}{2 + 2t}$$

$$= \int_0^1 \frac{dt}{1+t}$$

$$= [\ln|1+t|]_0^1$$

$$= \ln 2$$

1 mark.

$$\text{(ii) } I = \int_0^{\frac{\pi}{2}} \frac{x}{1 + \cos x + \sin x} dx$$

Let $u = \frac{\pi}{2} - x$

$\therefore du = -dx$

$$\therefore I = - \int_{\frac{\pi}{2}}^0 \frac{(\frac{\pi}{2} - u)}{1 + \cos(\frac{\pi}{2} - u) + \sin(\frac{\pi}{2} - u)} du$$

1 mark

$$= \int_0^{\frac{\pi}{2}} \frac{\frac{\pi}{2} - u}{1 + \sin u + \cos u} du$$

1 mark

$$\therefore I = \int_0^{\frac{\pi}{2}} \frac{\frac{\pi}{2}}{1 + \sin u + \cos u} du - I$$

1 mark.

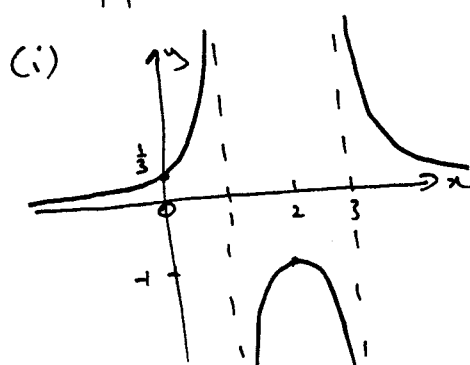
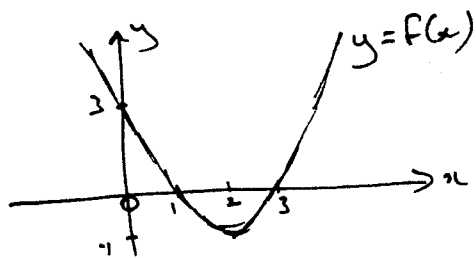
$$2I = \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \frac{du}{1 + \sin u + \cos u}$$

$$= \frac{\pi}{2} \ln 2$$

1 mark

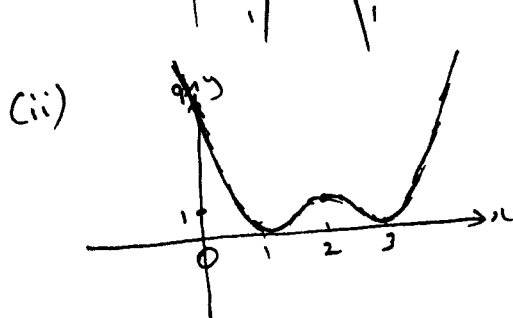
$$\therefore I = \frac{\pi}{4} \ln 2$$

c



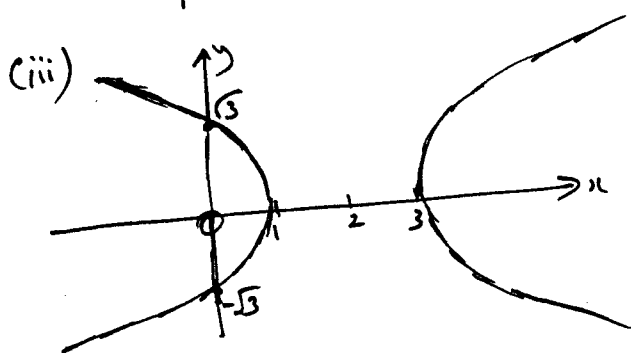
$$y = \frac{1}{f(x)}$$

1 mark.



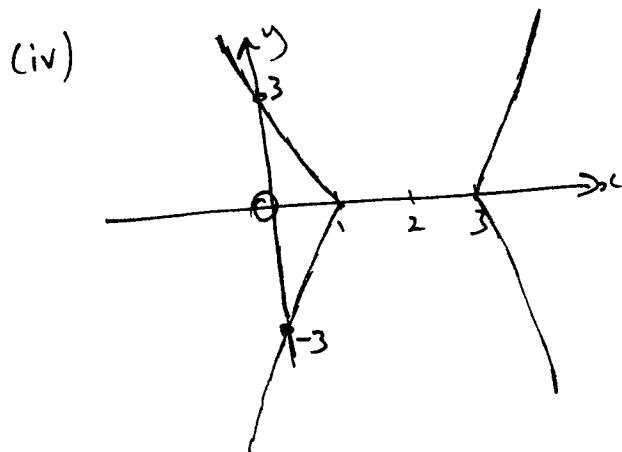
$$y = [f(x)]^2$$

2 marks



$$y^2 = f(x)$$

2 marks

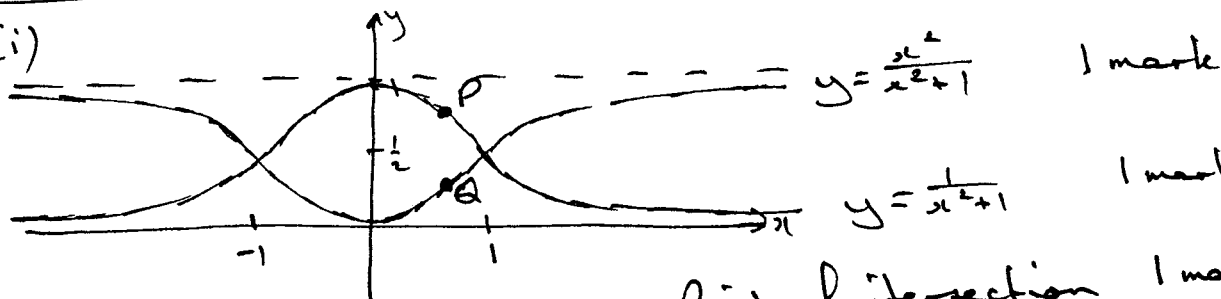


$$|y| = f(x)$$

1 mark.

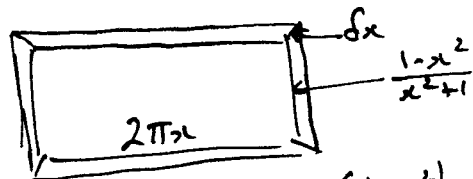
Question 7

a (i)



(ii) Take thin slice, through $P(x, y)$ on $y = \frac{x^2}{x^2+1}$ as shown, thickness δx . Rotate slice about y axis to form a thin-walled hollow cylindrical shell of: inner radius x
height $PQ = \frac{1}{x^2+1} - \frac{x^2}{x^2+1}$

Vol. of shell



$$\delta V = 2\pi x \left(\frac{1-x^2}{x^2+1} \right) \delta x$$

$$\text{Volume} = \lim_{\delta x \rightarrow 0} \sum_{x=0}^1 2\pi x \left(\frac{1-x^2}{x^2+1} \right) \delta x$$

$$= 2\pi \int_0^1 \frac{-x^3 + x}{x^2+1} dx$$

$$= 2\pi \int_0^1 \left(-x + \frac{2x}{x^2+1} \right) dx$$

$$= 2\pi \left[-\frac{x^2}{2} + \ln(x^2+1) \right]_0^1$$

$$= 2\pi (2\ln 2 - 1) \text{ cu. units}$$

b (i) $F_8 = F_7 + F_6$
 $= (F_6 + F_5) + (F_5 + F_4)$
 $= (F_5 + F_4) + 2F_5 + F_4$
 $= 3F_5 + 2F_4$

(ii) Prove true for $n=1$ $F_4 = 3$, which is div. by 3

Assume true for $n=k$

ie assume F_{4k} is div. by 3

ie assume $\exists$ integer M such that $F_{4k} = 3M$

Prove true for $n=k+1$ if true for $n=k$

ie prove F_{4k+4} is div. by 3 if F_{4k} is div. by 3

$$\text{Now } F_{4k+4} = F_{4k+3} + F_{4k+2}$$

$$\begin{aligned}
&= (F_{4k+2} + F_{4k+1}) + (F_{4k+1} + F_{4k}) \\
&= (F_{4k+1} + F_{4k}) + 2 \times F_{4k+1} + F_{4k} \\
&= 3 \times F_{4k+1} + 2 \times F_{4k} \\
&= 3 \times F_{4k+1} + 6M \\
&= 3(F_{4k+1} + 2M), \text{ which is div. by } 3 \quad 1 \text{ mark.} \\
&\quad \text{as } F_{4k+1} + 2M \text{ is an integer}
\end{aligned}$$

Conclusion

True for $n=k+1$ if true for $n=k$

But true for $n=1$

$\therefore$ true for all integer $n \geq 1$

1 mark.

Question 8

Q (i) $(\cos \theta + i \sin \theta)^3 = \cos 3\theta + i \sin 3\theta$

$$\therefore \cos^3 \theta + 3 \cos^2 \theta i \sin \theta + 3 \cos \theta i^2 \sin^2 \theta + i^3 \sin^3 \theta = \cos 3\theta + i \sin 3\theta$$

Equating real parts $\cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta$
 $= \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta)$
 $= 4 \cos^3 \theta - 3 \cos \theta$

1 mark.

(ii) $8x^3 - 6x + 1 = 0$
 $4x^3 - 3x = -\frac{1}{2}$

Let $x = \cos \theta$

Eqn becomes $4 \cos^3 \theta - 3 \cos \theta = -\frac{1}{2}$

$$\cos 3\theta = -\frac{1}{2}$$

$$3\theta = \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}$$

$$\theta = \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{8\pi}{9}$$

Hence $8x^3 - 6x + 1 = 0$ has solutions

$$\cos \frac{2\pi}{9}, \cos \frac{4\pi}{9}, \cos \frac{8\pi}{9}$$

1 mark.

(iii) Using prod. of roots $= -\frac{d}{a}$

$$\cos \frac{2\pi}{9} \cdot \cos \frac{4\pi}{9} \cdot \cos \frac{8\pi}{9} = -\frac{1}{8}$$

1 mark

$$\cos \frac{2\pi}{9} \cos \frac{4\pi}{9} (-\cos \frac{\pi}{9}) = -\frac{1}{8}$$

$$\therefore \sec \frac{\pi}{9} \sec \frac{2\pi}{9} \sec \frac{4\pi}{9} = 8$$

1 mark.

b (i) $x = \theta - \sin \theta$

$$y = 1 - \cos \theta$$

$$\frac{dx}{d\theta} = 1 - \cos \theta$$

$$\frac{dy}{d\theta} = \sin \theta$$

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

1 mark

$$= \frac{\sin \theta}{1 - \cos \theta}$$

let $t = \tan \frac{\theta}{2}$

$$= \frac{2t}{1+t^2} \cdot \frac{1-t^2}{1-t^2}$$

$$= \frac{2t}{2t^2}$$

$$= \frac{1}{t}$$

$$= \cot \frac{\theta}{2}$$

1 mark

$$\begin{aligned}
 \text{(ii)} \quad \frac{d^2y}{dx^2} &= \frac{d}{dx} \left(\cot \frac{\theta}{2} \right) \\
 &= \frac{d}{d\theta} \left(\cot \frac{\theta}{2} \right) \cdot \frac{d\theta}{dx} \\
 &= -\frac{1}{2} \operatorname{cosec}^2 \frac{\theta}{2} \times \frac{1}{1 - \cos \theta} \\
 &= \frac{-1}{2 \sin^2 \frac{\theta}{2} (1 - \cos \theta)} \\
 &= \frac{-1}{(1 - \cos \theta)(1 - \cos \theta)} \\
 &= -\frac{1}{y^2}
 \end{aligned}$$

1 mark

1 mark.

$$\begin{aligned}
 \text{(iii)} \quad \text{Stat. pts when } \frac{dy}{dx} &= 0 \\
 \cot \frac{\theta}{2} &= 0 \\
 \frac{\theta}{2} &= \frac{n\pi}{2}, \quad n \text{ odd} \\
 \therefore \theta &= n\pi, \quad n \text{ odd}
 \end{aligned}$$

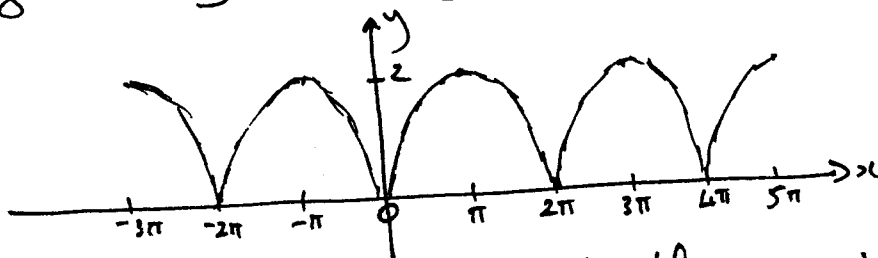
1 mark

$$\begin{aligned}
 \text{At these points, } x &= n\pi - \sin(n\pi) \\
 &= n\pi \\
 \text{and } y &= 1 - \cos(n\pi) \\
 &= 2
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Stat. pts at } (n\pi, 2) \quad &\text{for } n \text{ odd} \quad 1 \text{ mark.} \\
 \text{(iv)} \quad \text{Since } \frac{d^2y}{dx^2} &= -\frac{1}{y^2}, \text{ then for each stat. point, } \frac{d^2y}{dx^2} = -\frac{1}{4} \\
 \therefore \text{all are max. turning points} & \quad 1 \text{ mark.}
 \end{aligned}$$

$$\begin{aligned}
 \text{(v)} \quad \text{Cuts } x \text{ axis when } y &= 0 \\
 \therefore 1 - \cos \theta &= 0 \\
 \theta &= 0, \pm n\pi \text{ where } n \text{ even} \\
 \text{Range is } 0 \leq y \leq 2 \quad &(\text{as } y = 1 - \cos \theta \text{ and } -1 \leq \cos \theta \leq 1)
 \end{aligned}$$

1 mark



1 mark.

(vi) The points where the curve cuts the x axis are critical points, where $\frac{dy}{dx}$ is undefined and tangents are vertical.

1 mark